

Augmentation Residency: An Immersive Live-Work HCI R&D Model

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ABSTRACT

The growing affordability of emerging consumer devices and accessibility of software-hardware prototyping represents an opportunity for a new era of personal fabrication of augmentation technologies. In this case study, we review the inaugural Augmentation Residency, a live-in creative technology residency that implemented a new approach to the collaborative development of human-computer interaction (HCI) systems. Over the course of ten weeks, eight students and young professionals from a wide range of disciplines lived and worked together to build and self-experiment with technologies that integrate extended reality (XR), wearables, brain-computer interfaces (BCI), and artificial intelligence (AI). Made possible by a residential community and interdisciplinary teams, this high-intensity model for HCI research and development (R&D) was structured to encourage experimentation with atypical human-centered design methods and artistic practices. We present our program design, organization, and management, and discuss challenges and suggestions for improvements, to inspire future residencies as vehicles for intensive HCI innovation.

CCS CONCEPTS

• **Human-centered computing** → **HCI theory, concepts and models**; *HCI design and evaluation methods*; *Interactive systems and tools*; • **Social and professional topics** → *Informal education*.

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1 INTRODUCTION

1.1 The Residency Model for Innovation

The increasing accessibility of resources for hardware-software development is driving a new era of rapid prototyping and small-team innovation. Residencies have existed in the art world since the 16th century to promote innovation and skill-building, and have recently become a popular format for multidisciplinary, collaborative efforts between engineers and researchers (e.g. HIAP Residency Program [36]) [41]. However, while such art-science programs are valuable for mutual inspiration [14, 30, 42], they often lack orientation toward producing useful creations [39]. Universities and research organizations offer programs [1, 40, 46] to engage students in HCI research and development, but often do not enable students to explore projects of their own interest collaboratively. Startup incubators (e.g., Y Combinator [48], Techstars [44], buildspace [9]) and hacker houses (e.g., Arcadia House [2], Hacker fellows [23]) propose an increasingly popular avenue environment for independent and fast-paced innovation. However, their profit-driven nature and limited runtime typically do not allow for meaning-driven exploration of hardware-software HCI R&D.

To address this, we designed the 2023 Human Augmentation Residency [28], a 10-week long, fully immersive, and all-expenses-paid program in which eight¹ artists, engineers, and researchers lived and worked together to build technologies that enhance human capabilities and experiences. We drew from our experiences running the Augmentation Lab at Harvard, art residencies, policy organizations, startups, R&D projects, and learnings from existing models. The Residency took place over the summer (June 20 - August 28, 2023) and involved an interdisciplinary cohort with backgrounds in electrical engineering, computer science, industrial design, performance arts, neuroscience, physics, and human augmentation. Our participant cohort was supported by a network of advisors, equipment sponsored by various partners, and the entrepreneurial atmosphere of co-working spaces in San Francisco. Incorporating

¹Out of the eight residents, four of them were both organizers and participants of the program.



Figure 1: A collage of five images show participants using a variety of technologies. From left to right: Alice and Treyden experiment with the Magic Leap AR headset, Aedan solders a PCB, Dünya showcases the OpenBCI equipment, Aida sets up an EMG-EMS machine, and Inga checks voltages using a multimeter.

practices from research, engineering, startups, and artistic story-telling, the Residency provided a fast-paced living-working environment promoting close-knit teams to develop human augmentation technologies. Fig. 1 depicts some highlights from our program.

1.2 The Case for Residential Human Augmentation

Human Augmentation (from Latin *augere*: “to increase, make big, enlarge, enrich”) [18] seeks to extend or enhance human sensory, motor, or cognitive capacity through the careful design and application of technology. Modern technological augmentation systems include smartwatches, augmented reality (AR) glasses, neural implants, and AI applications for productivity enhancement. Converging advancements in generative AI tools, extended reality (XR) devices, brain-computer interface (BCI) devices, and wearable biosensors present new opportunities for people to connect and integrate these systems to produce their own augmentation technologies.

In combination with large-scale access to computers and the internet, open-source software [38], readily-modifiable hardware [47], and accessible digital fabrication with communities like Fab Labs [7], independent developers are now able to build and customize their own augmentation technologies more than ever before. Such consumer-facing technologies will only continue to mature and intersect, with products such as ChatGPT [49], Stable Diffusion [37], and Quest Pro headset released in the past year. It seems that we are at an inflection point for the independent and distributed development of human augmentation technology.

To this acceleration of consumer technologies, dissonant cultural responses have emerged. For example, transhumanists seek to evolve the human race beyond our physical limitations using technology [27], while technological skeptics advocate for a more natural way of life [29]. In our own reflections, we identified a similar dissonance within the HCI community between the ideal intended use of human augmentation technologies and their actual use in our own lives. We found ourselves asking: *As HCI developers, why don't most of us regularly use the technologies we develop?* A key motivation of our residential program was to encourage the development of augmentation technologies that we could live with to understand by direct experience their use, impact, and limitations.

In this emerging era of personal augmentation, the residential model brings technology development into contact with the living

values of the developers – through self-experimentation, intensive collaboration, and intimate discourse, the residential model functions as a vehicle to investigate the ethics and impact of human augmentation, its role in our present and future lives, and our responsibility as creators of human-interfacing technologies.

2 OBJECTIVES

The Residency was designed with an open-ended format, with a focus on process and adaptability over strict measurable outcomes. We designed this model using three major objectives:

- (1) **Interdisciplinary teams** to integrate creative and engineering backgrounds. Countless studies have shown the benefits of interdisciplinary collaboration in academia [13] and industry [8]. As Harris writes on Xerox PARC, “If you put creative people in a hothouse setting, innovation will naturally emerge” [22]. Thus, we curated a multidisciplinary cohort with expertise across a broad range of creative and technical fields.
- (2) **Residential community** to facilitate more intimate and intensive collaboration. A physical work environment supports creative work in three ways: functional, psychosocial, and inspirational [24]. Pool et al. observed residencies to be a critical method for pushing boundaries in innovation and thinking by establishing a story between the creation practice and a physical site [35]. Supporting more holistic insight and collaboration, we harnessed the proximity of the immersive live-work environment to promote the cross-pollination of ideas and the collision of different perspectives across the cohort.
- (3) **Human-centered design** to guide meaningful technology development. Technology encompasses both devices and the behaviors they afford. We took inspiration from human-centered design methodologies to encourage an understanding of the human behaviors elicited by each teams’ technical products. Methodologies included human-centered design [3], design thinking [45], emotion-centered design [17], and speculative design [19]. From these, we encouraged five distinct approaches to help scope and align project development during the Residency: 1) low-tech self-experimentation, 2) user research, 3) integration of narratives & artistic practice, 4) rapid prototyping, and 5) low-tech / high-tech parallel development.

In executing these objectives, we sought to 1) provide resources, mentorship, and community for the residential cohort, 2) produce HCI demos, research papers, and products as outputs, 3) explore philosophical and ethical perspectives on augmentation, and 4) present our ideas and work in a culminating showcase. Fig. 2 depicts the full organization process from initial setup to final showcase.

3 METHODS

3.1 Interdisciplinary Teams

3.1.1 Residents. To curate a balanced cohort, we sought applicants with creative breadth and technical skills across embedded systems, ML, BCI, and XR. Calls for applications were posted internationally in university clubs, research labs, maker groups, fellowships, and public online forums. We evaluated applicants based on alignment with our vision, technical background, emotional maturity in communication and living together, and the program's relevance to the applicant's career. This was achieved through reviewing their portfolios, written answers to questions, and remote 30-minute interviews. From over 170 applications, the four organizers interviewed 30 applicants over Zoom and ultimately selected four people, making up a total cohort of eight residents²: three college undergraduates, one recent college graduate, two masters students, one PhD student, and one industry professional.

3.1.2 Mentors. We curated a network of expert mentors with diverse perspectives, competencies, and life experience to provide feedback and help our teams form more holistic approaches to their work. We both reached out to and received offers for mentorship from professors, startup founders, professional researchers, and industry leaders. During ideation, building, and testing, six mentors provided technical project reviews through synchronous calls and asynchronous Discord communications. In-person mentors also led discussions of ethics, philosophy, and social impact over dinners.

3.1.3 Team & Project Formation. In initial onboarding sessions, we utilized chaotic brainstorming techniques to imagine ideas freely and outrageously, collating over 100 ideas. Using Padlet bulletin board software [34], we clustered our ideas into three categories: Senses & Emotions, Mind & Control, and Storytelling & Creative Practices, from which preliminary projects and collaborations emerged. Participants then engaged in a feedback loop between individual research and conversations with people with shared interests, during which groups that did not converge on an idea voluntarily dissolved, and new groups formed dynamically.

3.2 Residential Model

3.2.1 Funding & Resource Acquisition. To acquire funding and resources for the Residency, we emailed over 300 companies, funds, and individuals in augmentation-related fields (e.g. XR, biotech, AI, wearables). We pitched our vision, prior successes with R&D, and sponsorship package, which offered publicity and access to our network of talent. Direct outreach proved effective for in-kind sponsorships of hardware (BCI headsets, XR headsets, 3D printers, etc.). We found monetary support through personal referrals to companies and private donors. Ultimately, the program cost around

\$40k, including \$4k per resident for 2.5 months of food and housing, as well as \$8k to fund all projects.

3.2.2 Location & Residential Makerspace. We chose San Francisco for its vibrant builder community, and selected an apartment that balanced cohort safety, cost of rent, and proximity to nearby makerspaces. In addition to using local makerspaces, we facilitated all-hours prototyping by building a makerspace in the apartment's living room with essential equipment that included a 3D printer, soldering station, oscilloscope, power supply, BCI headsets, XR headsets, and wearable devices such as haptic gloves and smart-watches.

3.2.3 Cohabitation. We developed and adapted systems of living together with the goal of promoting physical and emotional well-being. Initially, we cooked and ate together. Later, to better support a healthy diet and minimize meal-planning effort, we switched to a group meal plan that began the third week. While cooking together facilitated closer personal relationships and a sense of community, the meal plan saved significant time for work. We also promoted a culture of open communication, initially implementing a morning work & life check-in during which residents were encouraged to ask for help and give feedback on the program. Over time, the sharing process became less formalized, with concerns discussed in the group as they arose. We also organized field trips to comedy shows, theaters, and more, encouraging creative interactions and a balanced lifestyle.

3.2.4 Community Culture. Fostering a sense of community, residents frequently explored personal and philosophical questions about life, our relationship with technology, and the future of humanity over dinner. We also watched sci-fi movies and engaged in creative practices such as dancing and acting together. Outside of the house, we visited Zen temples, comedy shows, parks, and participated in seminars, hackathons, and workshops. These collective activities often provided new insights and sometimes even directly contributed to the development of our technologies.

3.3 Human-Centered Methodologies

3.3.1 Low-tech self-experimentation to define project directions. We explored a problem-focused, technology-last approach in which low-tech self-experimentation was used to identify project directions that were personally relevant and meaningful to each developer. This process involved 1) introspection and discussion to reveal the latent needs for augmentation in our own lives (e.g. personal change, interpersonal relationships), 2) brainstorming of low-tech methods that could achieve augmentation (e.g. writing, role-play), and 3) self-experimentation and iteration with low-tech methods that informed the development of higher-tech final outputs.

3.3.2 User research. Drawing from HCD and design thinking approaches [20], teams conducted user research via semi-structured interviews and public surveys to explore diverse outside perspectives. Insights from user research guided systematic identification of user segments and helped define user needs.

3.3.3 Integration of narratives & creative practices. Artists and storytellers are directed to seek truth, to question assumptions, and to discover and understand meaning and the human condition. In

²Based on our prior experience, we considered a cohort of six to ten the ideal size for a close-knit group in which each person could work on multiple projects.

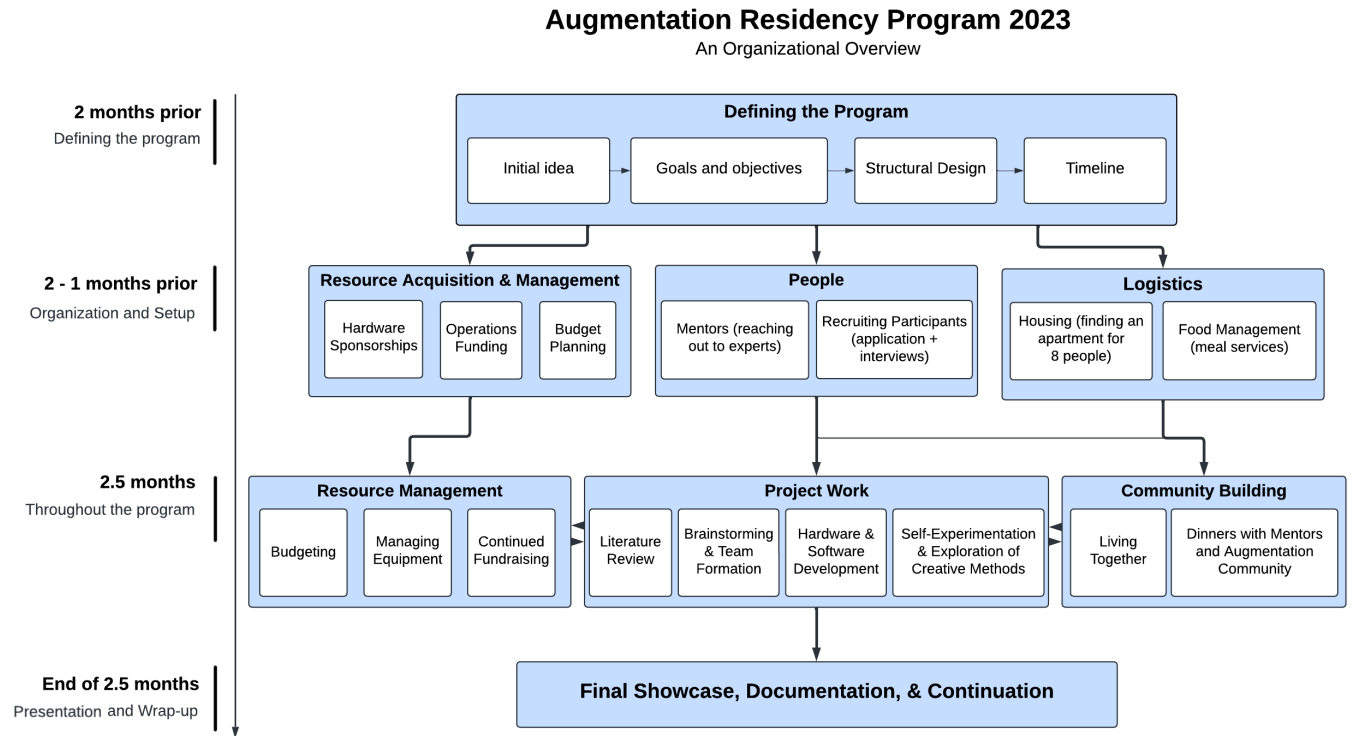


Figure 2: Chart depicting the Residency organization timeline, starting from defining the program two months prior to the organizational stages to management during the program to the Final Showcase.

order to deepen holistic understanding of problem areas and create solutions that engage with experiential layers of meaning, we integrated narratives and creative practices into the tech development process. Creative practices, such as writing, movement, acting, and play, were investigated as low-tech augmentations to human behavior that could be enhanced and guided by technology. These practices were leveraged to inspire new ideas, inform development, and in some projects developed into integral parts of the final product (e.g. as interactive media, product trailers, user journey narratives).

3.3.4 Rapid prototyping for self-experimentation. Following the ready-fire-aim methodology [21], hardware projects focused on rapid prototyping and iterative testing in daily life. This enabled a faster understanding of user experiences, adaptation to user needs, and refinement of project goals based on emergent insights from self-experimentation.

3.3.5 Low-tech / High-tech parallel development. Low-tech augmentations (e.g. non-electronic devices) can often feel more natural and be faster to prototype, while high-tech augmentations can be more complex and adaptive at the cost of slower development timelines. A two-pronged competitive approach in which both low-tech and high-tech variations are developed in parallel can reveal trade-offs and provide more holistic understanding of the solution space.

4 RESULTS

The Residency produced 1) eight HCI projects (depicted in Fig. 3) integrating BCI, XR, AI, biosensing, and haptics for sensory, emotional, interpersonal, physical, and creative augmentation, 2) comparative insights on human-centered methodologies, 3) a collectively-produced philosophical zine, and 4) a culminating showcase.

4.1 Project Outputs

- **Haptic Ring** [50] built a sensory augmentation device for acquiring new senses via vibration. The team designed and fabricated the electrical and mechanical systems for their first Minimum Viable Product (MVP), which translates magnetometer data into vibrational patterns. Post-residency, they received an innovation grant and are writing the firmware for the system.
- **Imagination Engine** [25] trained a machine learning model to translate brain activity into abstract art, with the vision of exploring the intricate language of emotions elicited by art encoded within EEG data.
- **Enact** [6, 16] developed a web application that uses role-play and AI-driven memory augmentation for resolving interpersonal relationships and practicing difficult conversations. Post-residency, the team is continuing to expand the product as a startup as part of the Harvard Innovation Lab.

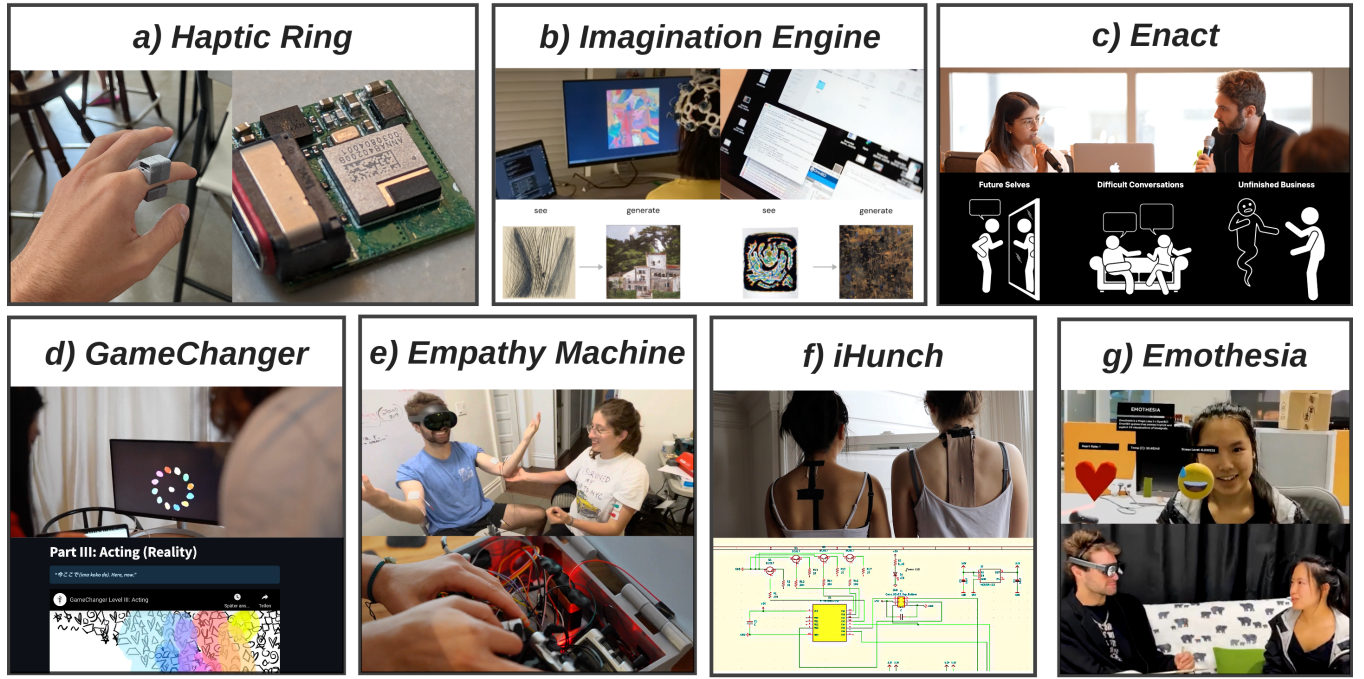


Figure 3: Images of the eight HCI projects. a) The *Haptic Ring* in assembled form and one of its PCBs, b) training the *Imagination Engine*, and examples of the output, c) two people showcasing *Enact* and graphics of the project’s use cases, d) *GameChanger*’s animation and website. e) the *Empathy Machine* tested on two people and its hardware components, f) two people testing the low-tech versions of *iHunch* and *iHunch*’s high-tech PCB schematic, g) *Emothesia*, visualizing heart rate, temperature, and stress level in an AR headset.

- **GameChanger** [10] made an experimental narrative-based game that helps people change. The team developed an initial web-based prototype of the system that integrates animations, interactive texts, guided meditations, and Fluxus-inspired event scores.³ Post-residency, the team is exploring new mediums for the game.
- **Empathy Machine** [5] aimed to simulate the feeling of touch-mirror synesthesia between two people through sensing and stimulating muscle activity. The team developed iterations of the hardware system and experimentally explored the dynamics of empathy and power created through this enforcement of body-mirroring.
- **iHunch** [4] explored low-tech and high-tech systems that help people correct bad posture through tactile feedback. The team designed a low-tech elastic posture tape prototype and a high-tech electronic prototype.
- **Emothesia** [11] developed a wearable system for visual-audio-haptic mapping of biosignals to enhance human interoception of our bodies and emotions. The headset-watch system helps two people sense and sync heart beats, and visualize heart rate, temperature, and electrodermal activity (EDA) signals in AR.

- **MIT Media Lab Collaboration.** Part of our cohort collaborated with researchers at the MIT Fluid Interfaces group to develop methods for reconstructing visual imagery from EEG data.

4.2 Methodologies in Action: Projects as Mini-Case Studies

4.2.1 Low-tech self-experimentation to define project directions. In *Enact*, we experimented with partner role-play methods for dealing with difficulties in interpersonal relationships, drawing inspiration from role-play methods in therapy [31]. For instance, we would act out emotional situations in fictional and fantasy settings or role-play as a real person from someone’s life to practice difficult conversations and find emotional closure. We iterated the experiments based on these findings, progressively integrating more technology to address challenges. For example, experimentation revealed that AI-generated scripts help overcome the effort in imagining our own fictional settings, which informed the development of our final application.

4.2.2 Integration of narratives & creative practices. The development of *GameChanger* relied on the creation of the central narrative about the player’s journey of change. We developed the game’s narrative through creative practices (e.g. writing and sketching) and conversations that investigated our personal experiences with change. Integrating insights from these explorations with existing

³Fluxus was an avant-garde art movement emphasizing fluid creativity and self-reflection[33]. Fluxus scores are event scores with brief, often playful instructions or proposals for actions that blend performance art and life.

theories in Jungian psychology, cognitive neuroscience, and behavioral science, we developed a series of animations, interactive texts, guided meditations, and Fluxus-inspired event scores that guide the player through a three-phase story of change: I) Seeing (Abyss), in which players confront their shadows [26], II) Imagining (Multiverse), in which players imagine all the possible selves they could be, and III) Acting (Reality) in which players enact their changed identities through living [32].

4.2.3 Rapid prototyping for self-experimentation. In *Haptic Ring*, we focused on rapid technical design and implementation, testing and iterating through multiple mechanical prototypes to find a flexible one-size-fits-all ring shape that could accommodate many different finger sizes. Various cohort members volunteered to each wear the ring for a day. This revealed different user experiences, including situations in which the Ring proved inconvenient or uncomfortable.

In *Empathy Machine*, we rapidly prototyped a system using EMG sensors and an off-the-shelf TENS unit that allowed us to experiment with different body-control games. This hands-on experimentation revealed insights about the system's potential for creating different, even challenging, dynamics between people. These insights were explored in further iterations.

4.2.4 Low-tech / High-tech parallel development. *IHunch* investigated contrasting approaches to a central problem – one teammate developed a low-tech system using only mechanical mechanisms and material properties, while the other teammate developed a high-tech system using custom electronics. This parallel development process facilitated discourse and competition between the two teammates towards ultimately more holistic understandings and solutions.

4.3 Culmination: Showcase and Zine

To share our work, we designed the Augmenting Evolution Showcase (depicted in Fig. 4) as the public-facing culmination of our residency program. The two-hour showcase involved an introduction of the Residency, project presentations, and unstructured time for guests to visit project booths, experience live demonstrations, and engage in conversations. The showcase was accompanied by the *Augmentation as (a lens for) living zine* [12] that emerged from group discussions about our relationship with technology and the meaning of augmentation.

5 DISCUSSION

5.1 Logistics & Organization

5.1.1 Choosing the right people. We prioritized technical skill in our recruitment process, but in hindsight, we realize that value alignment, motivation, and personal fit are more significant factors in determining cohort dynamics and overall program success. Differences in values or motivations are difficult to assess in interviews and can often create conflict or divergence within newly-formed teams, especially in an intensive residential format. Therefore, we suggest a referral-based approach for programs of this intensity and intimacy to mitigate uncertainty in personal compatibility.

5.1.2 Simultaneously organizing and participating. In self-organized programs, organizers who also participate face several unique challenges. First, there can be imbalanced dynamics between residents who are part of the organizing team and those who are not. To restore a shared sense of ownership across the cohort, we recommend requiring all residents to take on responsibilities. Second, administrative work takes significant time and mental focus away from participation in projects. Setting regular meeting times with time caps helped our organizing team mitigate this. Finally, agreeing on a managerial approach for the program is essential: some organizers preferred *laissez-faire* and others preferred a more structured approach. From our experience, residencies that focus on team work with predefined outputs (e.g. a showcase) would benefit from more structure to ensure the cohesive teams and consistent progress. In contrast, a residency in which participants work independently may find more benefit from a *laissez-faire* approach.

5.1.3 Collaborating effectively. Intimate collaborations present both synergies and challenges to interpersonal coordination, trust, and conflict. Within the residential format, successful collaboration required that teams reconcile conflicting preferences on work locations (e.g. at home environments, coffee shops, or makerspaces) and work times. Complexity also arose from participants juggling multiple projects, with different collaborators and moving at different paces. Initially, a lack of clear communication about priorities, expectations, and contributions among team members led to tensions and conflicts. The improvisational approach of artistic and self-experimental methodologies exacerbated ambiguities in teamwork. For future programs, we recommend more structured approaches to facilitating collaboration, as well as encourage open and honest communication about roles and commitment to ensure clarity on progress, goals, and equitable contributions from each team member.

5.2 Integration of Work & Life

5.2.1 Finding work-life balance and boundaries. Managing work-life boundaries and the nuances of work-friendship relationships proved challenging. The lines between personal and professional interactions are easily blurred when living and working together, especially while engaging in self-experiments and personal practices as part of work. This occasionally led to emotionally heightened conditions, exacerbated by limited private space. It also underscored the importance of outside perspectives, recesses, and maintaining a balance between immersion and external exploration. We recommend setting aside time for recreational activities, such as communal dinners or exercise, to establish a more explicit boundary between the professional and personal context. We also recommend setting community values with the cohort at the beginning of the program to help promote healthy interpersonal dynamics.

5.2.2 Experimenting in real life. The emotional intensity of the Residency was heightened by frequent self-experimentation on topics of human behavior, in which we intentionally put ourselves into vulnerable situations. We purposefully explored topics such as interpersonal trust, intimacy, and vulnerability to challenge our world views and constructions of the self. However, as these experiments were motivated by subjective exploration, they lacked a



Figure 4: Pictures from the Augmentation Showcase. First row: Residency cohort introducing themselves during the Augmentation Showcase. Second row from left to right: Augmentation Zine, *Haptic Ring* worn by Inga, Jennifer showcasing the *Imagination Engine*.

framework for scientific repeatability and safe boundaries. While many experiments proved eye-opening and beneficial for long-term personal growth, as well as insightful for need-based HCI development, we experienced some unanticipated adverse emotional effects in our personal lives and relationships due to the provocative nature of some of our self-experiments. Therefore, we recommend clearly defining the experimental context and interleaving behavioral experiments with other development methods unrelated to self-experimentation to allow time for emotional recovery.

5.3 Approaches to Work

5.3.1 Choosing what to work on. The autonomy to select a project focus and its outputs illuminated diverse philosophies and perspectives on technology development and its purpose. A core tension emerged between the human-first approach that prioritizes direct utility to a problem, and the tech-first approach that prioritizes the exploration of cutting-edge technological frontiers. Our initial vision focused on building technologies we could immediately use, but this constraint proved difficult with emerging technologies like XR, in which hardware is not yet sufficiently accessible. Ultimately, teams took a variety of approaches based on personal values: some focused on tackling technical challenges, some on finding real customers, some on producing artistic meaning. From this, we learned that teams need to agree on a project's fundamental motivation early on; otherwise, misalignment could stymie building or lead to conflicts down the line. We recommend future programs facilitate early and active discussion of philosophical motivations in developing technologies in order to help participants choose their focus and form fundamentally aligned teams.

5.3.2 Balancing structure and improvisation. While developing our projects, we experienced a trade-off between following a predefined project plan and improvising with the organic emergence of new directions during development. We contrast the *Haptic Ring* and *GameChanger* projects as example: the *Haptic Ring* project determined the topic in the first week of the Residency and followed a preplanned development timeline without pivoting, while the *GameChanger* project emerged in the fifth week from a series of prior explorations and self-experiments, and continued to evolve in medium and implementation through the end of the Residency. The improvisational approach was required by the objectives of *GameChanger*, as its purpose and meaning underwent continual rediscovery. Regardless of their development strategy, however, all projects found benefit from rapid prototyping. We strongly recommend the "ready, fire, aim" method to encourage reflection and iteration in design. This method mirrors the iterative element of the agile methodology [15]. We encouraged participants to rapidly produce "Little Bets" [43], which are small-scale experimental versions of ideas with low failure costs for fast testing, failing, and learning.

5.3.3 Finding dissonance and meaning in work. The structures dividing work and life usually shelter us from their often conflicting values. During the Residency, in contrast, we were forced to face any dissonance between our theoretical values in work and the values we enact in our real lives. The residential format prompted us to question why, for example, we often preferred old-fashioned or low-tech ways of living despite our commitment to developing new technologies. In attempting to live with the technologies that we built, we often discovered that augmentations we had imagined would be exciting (e.g. interpersonal biosignal monitoring) were

not, in actual experience, fulfilling. In integrating self-experimental and artistic practices into the development process, we sought to find personal meaning in work beyond the traditionally limited scope of technological rationales. We found this kind of holistic, immersive, and experiential HCI R&D conducive to both personal growth and more human-centric development of technology.

6 CONCLUSION

The 2023 Human Augmentation Residency was the inaugural 10-week technology development program organized by the Augmentation Lab. With a high-intensity residential model for HCI R&D, we sought to support a human-centered approach to innovation, bridging interdisciplinary teams and communal development to produce meaningful human augmentation technology and holistically informed work. In planning and executing our Residency, we discovered key challenges and opportunities to support for the future of such HCI R&D programs, including approaches to support work-life integration, setting boundaries in collaborative self-experimentation, and aligning on philosophical motivations for work. Our experiences with integrating living and working demonstrate the potential of residential models to facilitate meaningful, collaborative HCI development. On the theoretical side, our deliberations on the purpose of our creations revealed the need to develop a framework to support sustainable human-centric technology. We hope our insights and recommendations will be useful to HCI researchers exploring new experimental and creative development models, as well as organizers of future HCI R&D programs.

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